

NeuPIMs: NPU-PIM Heterogeneous Acceleration for Batched LLM Inferencing

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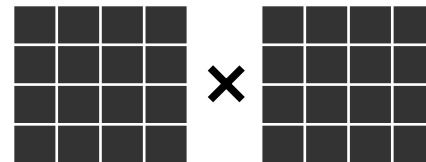


ASPLOS 2024



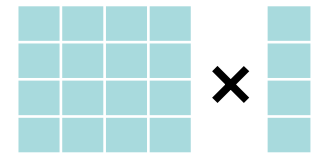
LLM batched inference
comprises **GEMM** and **GEMV**

GEMM



matrix-matrix
multiplication

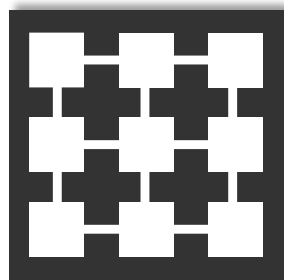
GEMV



matrix-vector
multiplication

GEMM

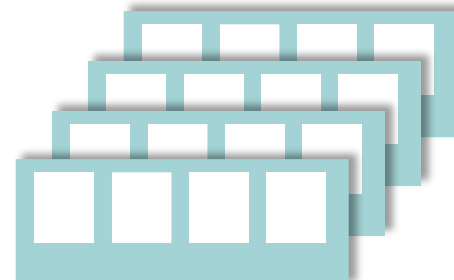
- compute-intensive
- well-suited to NPU



NPU

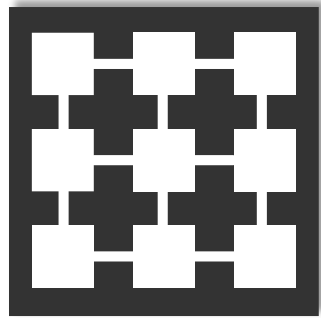
GEMV

- bandwidth-intensive
- well-suited to PIM

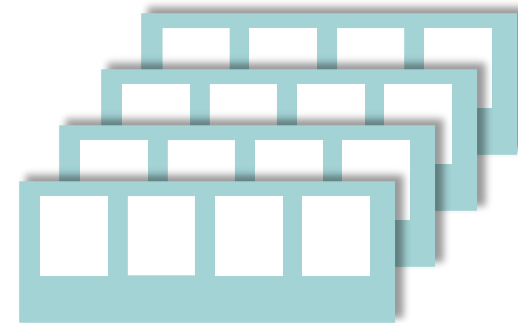


PIM

However, with naïve NPU+PIM integration,
system suffers from resource **underutilization**



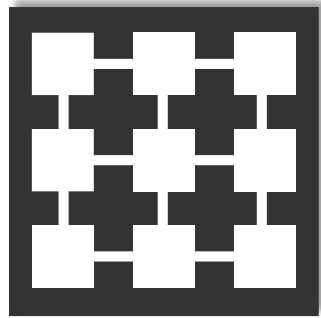
NPU



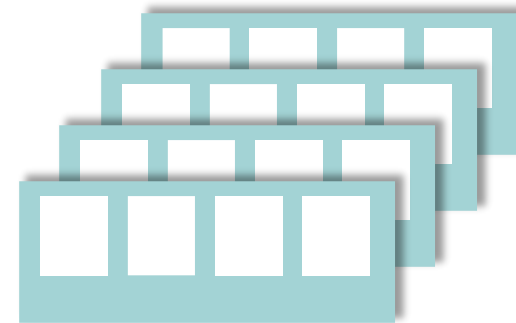
PIM



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NPU

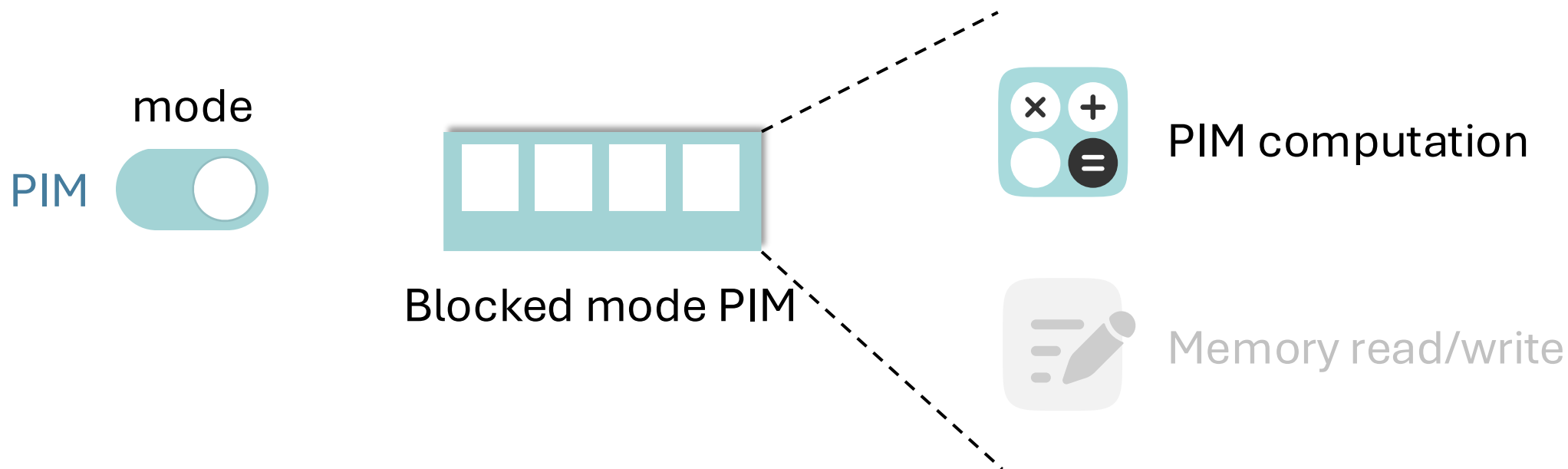


PIM



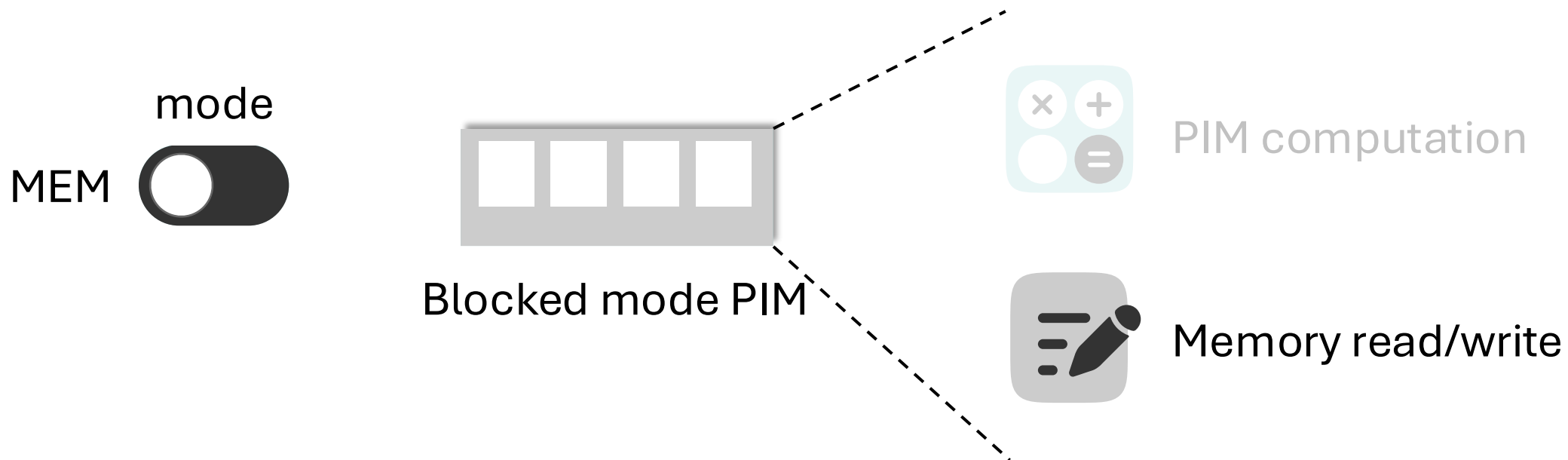
Challenge #1

Existing PIM operates in “blocked” mode



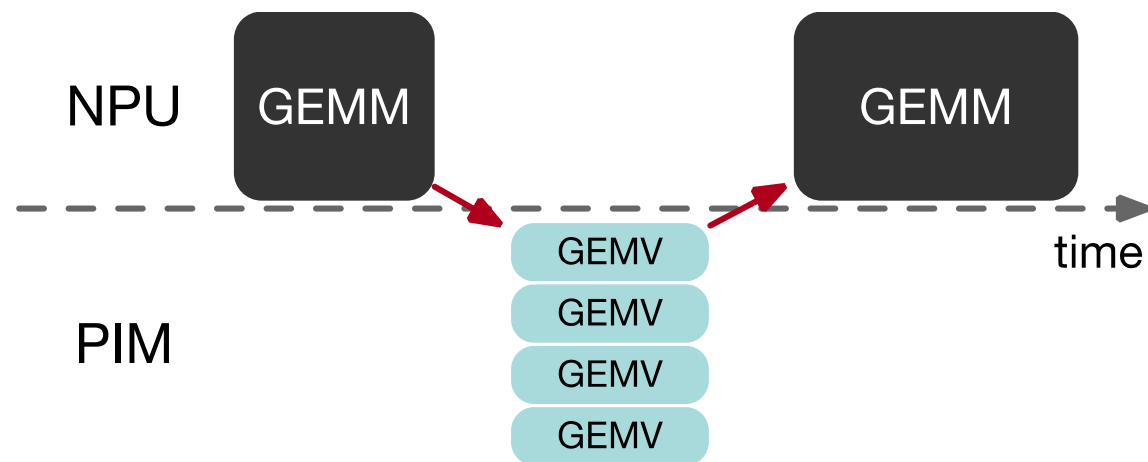
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Existing PIM operates in “blocked” mode

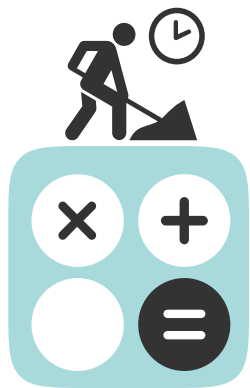


Challenge #2

GEMM and GEMV have algorithmic dependency in LLM



We devise **NeuPIMs**, NPU-PIM heterogeneous acceleration solution for batched LLM inferencing

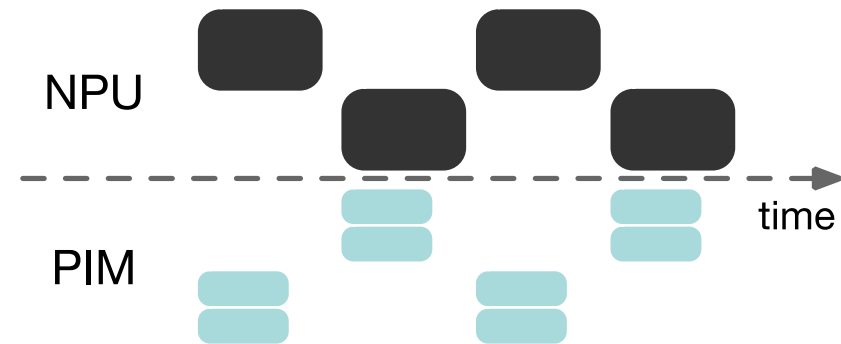


PIM



MEM

PIM with dual row buffers



Sub-batch interleaving

NeuPIMs: NPU-PIM Heterogeneous Acceleration for Batched LLM Inference

Session 6B

04/30 (Tue)

14:30

2.4×

throughput improvement
over NPU

1.6×

throughput improvement
over naïve NPU+PIM



Our simulator code is available
<https://github.com/casys-kaist/NeuPIMs>

